

TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

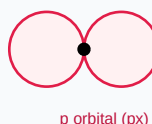
CHAPTER 12: BASIC RULES OF ORGANIC REACTIONS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Basic Rules of Organic Reactions

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

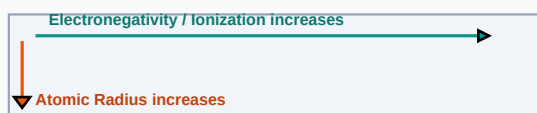
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Electrophile	An electron-deficient species that accepts electron pairs in chemical reactions, acting as a Lewis acid.
Nucleophile	An electron-rich species that donates electron pairs in chemical reactions, acting as a Lewis base.
Inductive Effect	The permanent polarization of a covalent bond caused by electronegativity differences along a carbon chain.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

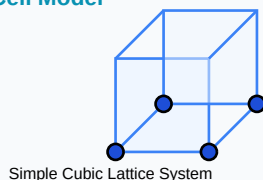
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Stability: Tertiary carbocation > Secondary > Primary > Methyl
- Inductive effect: +I (electron release: $-\text{CH}_3$), -I (electron pull: $-\text{Cl}$, $-\text{NO}_2$)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

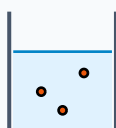
Example 1: Classify the following as electrophiles or nucleophiles: H^+ , OH^- , BF_3 , NH_3 .

Step-by-step Solution:

Electrophiles (electron acceptors): H^+ , BF_3 .

Nucleophiles (electron donors): OH^- , NH_3 .

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

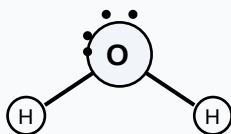
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain Electromeric effect, Resonance effect, and Hyperconjugation stability of alkyl radicals.

Analytical Solution Strategy:

- (1) Electromeric: temporary polarization of multiple bonds on demand of reagent.
- (2) Resonance: delocalization of pi electrons in conjugated systems.
- (3) Hyperconjugation: no-bond resonance involving sigma C-H electrons and empty p orbitals, stabilizing carbocations.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

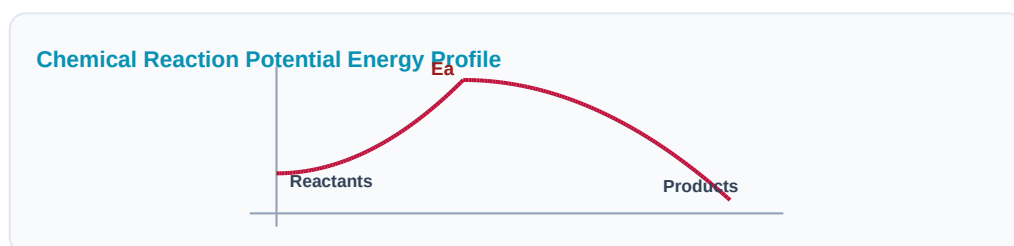
Q1 (1-Mark): Differentiate between homolytic and heterolytic bond cleavages.

Q2 (2-Mark): Explain inductive effect with an example.

Q3 (2-Mark): Why is tertiary carbocation highly stable?

Q4 (5-Mark): What is a free radical? Describe its hybridization.

Q5 (5-Mark): Compare nucleophilicity and basicity.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Polymer Chemistry

Free radical initiators open double bonds in monomers to manufacture plastics.

Application Case: Synthetic Biology

Nucleophilic attacks of amine residues link peptide chains in synthetic insulin.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

